

Research Project Description

Low-Latency Tri-Attention Transfer U-Net Framework for Robust Lung Segmentation in Chest X-rays Under Domain Shift

1. Project Summary

This project develops a deployment-oriented Tri-Attention Transfer U-Net framework for lung-field segmentation in chest X-ray images. The main goal is to improve boundary fidelity under weak pleural contrast, rib and clavicle overlap, heart-diaphragm interference, portable imaging artifacts, and cross-dataset domain shift, while also reporting real-time deployment feasibility. The framework is evaluated on three public datasets: Montgomery County, Shenzhen, and Darwin.

2. Problem Addressed

- Weak lung boundaries and low pleural contrast can cause leakage around the apices and costophrenic angles.
- Ribs, clavicles, heart shadow, diaphragm overlap, and portable artifacts can produce boundary rupture and under-segmentation.
- Models with high Dice scores often omit deployment indicators such as memory use, parameter count, model size, and inference time.
- Cross-dataset robustness remains difficult because CXR distributions vary across institutions, acquisition devices, and patient populations.

3. Proposed Method

The proposed model integrates a ResNet50 transfer-learning encoder with U-Net-style skip connections and a sequential tri-attention mechanism. Channel attention uses SE and ECA-style recalibration to emphasize informative feature channels; spatial attention focuses the decoder on lung-relevant regions; and criss-cross contextual refinement improves long-range structural consistency without the cost of full self-attention. At the bottleneck, ASPP captures multi-scale context, and decoder blocks recover fine lung boundaries through attention-filtered skip fusion.

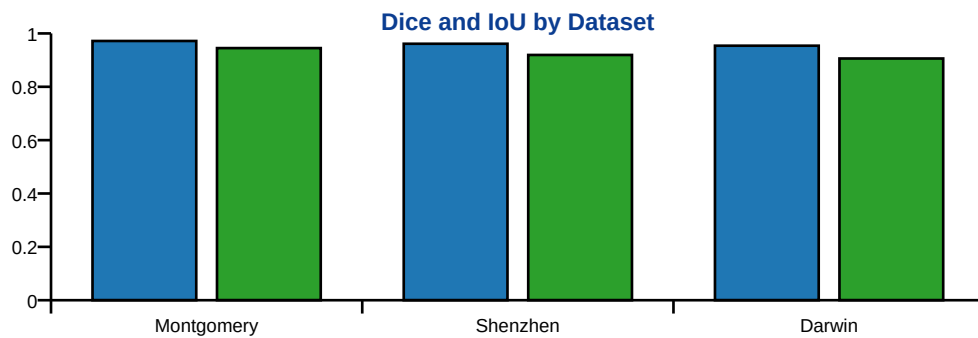
Component	Role in the Framework
ResNet50 encoder	Extracts hierarchical low-, mid-, and high-level chest features using transfer learning.
ASPP bottleneck	Aggregates multi-scale context without additional resolution loss.
Channel attention	Reweights informative channels and suppresses noisy/background channels.
Spatial attention	Improves localization of lung regions and boundary-relevant structures.
Criss-cross attention	Models long-range row/column dependencies for structural consistency.
Sigmoid output head	Produces a single-channel lung probability map for binary mask generation.

4. Experimental Workflow

Stage	Description
Data preparation	CXR images and masks are paired by filename, loaded in grayscale, resized to 128 x 128, normalized to [0, 1], and masks are binarized.
Training setup	A 70/15/15 train-validation-test split is used; Montgomery also uses 5-fold cross-validation because of its small sample size.
Optimization	Training uses a combined Dice, focal, and Tversky loss to improve overlap, foreground detection, and boundary sensitivity.
Evaluation	Dice, IoU, precision, sensitivity, specificity, AUC, NSD, MSE, PSNR, threshold calibration, error maps, and PR/ROC curves are reported.
Interpretability	Attention maps, Grad-CAM, Grad-CAM++, saliency maps, and spatial error maps examine anatomical relevance.

5. Key Quantitative Outcomes

Dataset	Dice	IoU	AUC	NSD	Inference Time
Montgomery	0.9717	0.9450	0.9934	0.982	4.40 ms
Shenzhen	0.9613	0.9193	0.9925	0.9632	10.26 ms
Darwin	0.9540	0.9060	0.9910	0.929	8.86 ms



6. Deployment and Clinical Relevance

The framework reports 41.53 million parameters, 158.41 MB memory footprint, and 4.40-10.26 ms inference per image. These results suggest low-latency feasibility for optimized edge or point-of-care workflows, especially if combined with mixed-precision inference, pruning, or quantization-aware training. The interpretability outputs indicate that the model generally focuses on anatomically meaningful lung regions rather than irrelevant background artifacts.

7. Limitations and Future Scope

The main limitations are dataset heterogeneity, reduced robustness on artifact-prone portable images, possible threshold variation across datasets, and limited validation outside the three public sources. Future work should include domain adaptation, pediatric and multi-institutional evaluation, uncertainty-aware quality control, and full end-to-end deployment timing including preprocessing and I/O.

Source Note

Prepared from the attached manuscript: Tri_Attention_UNET_28_May_2026_v9.pdf. Dataset and metric values should be verified against the final accepted manuscript before public release.